Module 14) Python – Collections, functions and Modules

- ## Accessing List

A list is an ordered, mutable collection of elements that can contain items of different data types. Lists are one of the most versatile data structures in programming.

Creation Methods

1. **Literal Syntax**: Created by enclosing elements in square brackets []
   - Empty list: []
   - Populated list: [element1, element2, element3]
2. **Constructor Method**: Using the list() function
   - Can convert other iterables (strings, tuples, etc.) into lists
3. **Comprehensions**: Compact way to create lists based on existing iterables
   - Follows the pattern: [expression for item in iterable if condition]

Element Access

Indexing Theory

- **Zero-based**: First element is at index 0
- **Negative Indices**: -1 refers to last element, -2 second last, etc.
- **Bounds**: Accessing beyond valid indices raises an IndexError

Nested List Access

- For multidimensional lists, use multiple indices: list[i][j]

- Each index operation accesses one level of nesting

- ## Indexing in lists (positive and negative indexing).

List indexing is the method of accessing individual elements within a list by their position. Python provides two types of indexing

1. **Positive Indexing** (left to right, starting at 0)
2. **Negative Indexing** (right to left, starting at -1)

Positive Indexing

- Begins at **0** for the first element
- Increases by 1 for each subsequent element
- The last element is at index len(list) - 1

Negative Indexing

- Begins at **-1** for the last element
- Decreases by 1 moving left through the list
- The first element is at index -len(list)

Comparison Table

Index	Positive	Negative	Element
1st	0	-5	'apple'
2nd	1	-4	'banana'
3rd	2	-3	'cherry'

Index	Positive	Negative	Element
4th	3	-2	'date'
5th	4	-1	'elderberry'

- ## Slicing a list: accessing a range of elements.

List slicing allows you to extract a portion (sub-list) of a list by specifying a range of indices. The basic syntax is:

list[start:stop:step]

Components of Slicing

1. Start Index (inclusive)
    - Beginning of the slice (default: 0)
    - If negative, counts from the end of the list
2. Stop Index (exclusive)
    - End of the slice (default: end of list)
    - The element at this index is not included
3. Step (optional)
    - Interval between elements (default: 1)
    - Can be negative to reverse the order